

BACHELOR THESIS
Dion Osmani

Optimisation de la planification énergétique urbaine par l'orchestration de l'IA

Informatique et Systèmes de communication (ISC)

Dion Osmani

Optimisation de la planification énergétique urbaine par l'orchestration de l'IA

Thesis submitted to the Bachelor degree programme

Informatique et Systèmes de communication (ISC) – Data engineering major

Haute École d'Ingénierie de Sion

Supervisor: **PROF. JESSEN PAGE**

Co-supervisor: **FLORIAN DESMONS**

Expert: **NILS SCHÜLER**

Submitted on: June 24, 2025

Abstract

The abstract of a bachelor thesis should provide a concise summary of the entire work. It typically includes:

- The context and motivation for the research.
- The main objective or research question.
- A brief description of the methodology or approach used.
- The key results or findings.
- The main conclusion or implications of the work.

The abstract should be self-contained, clear, and usually does not exceed 250–300 words. It allows readers to quickly understand the purpose and outcomes of the thesis without reading the full document.

The abstract **must** be written in both French and English.

Please also insert your project git/github URL HERE if your project is not confidential.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius.

Keywords : engineering, data, large language models, AI agents, energy planning

Résumé

Le résumé d'un mémoire de bachelor doit fournir un aperçu concis de l'ensemble du travail. Il inclut généralement :

- Le contexte et la motivation de la recherche.
- L'objectif principal ou la question de recherche.
- Une brève description de la méthodologie ou de l'approche utilisée.
- Les principaux résultats ou découvertes.
- La conclusion principale ou les implications du travail.

Le résumé doit être autonome, clair et ne pas dépasser habituellement 250 à 300 mots. Il permet aux lecteurs de comprendre rapidement le but et les résultats du mémoire sans lire l'intégralité du document.

Le résumé doit être rédigé en français **et** en anglais.

Veuillez également ajouter l'URL de votre git/github ici si le projet n'est pas confidentiel.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Quia ipsum suspendisse ultrices gravida. Risus commodo viverra maecenas accumsan. Ut enim aequo animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius.

Keywords : engineering, data, large language models, AI agents, energy planning

Acknowledgements

The **Acknowledgements** section of a bachelor thesis is where you express gratitude to those who supported you during your research and writing process. It is an **OPTIONAL** section. It may include:

- Academic supervisors or advisors who provided guidance.
- Professors or instructors who offered feedback or resources.
- Family and friends for emotional or practical support.
- Institutions or organizations that provided funding, facilities, or data.
- Anyone else who contributed significantly to your work.

Keep this section concise and sincere. It is typically placed after the abstract and before the main content of your thesis.

Table of Contents

1 – Writing a thesis	13
1.1 – The content of a thesis	13
2 – Introduction	15
3 – State of the Art	17
4 – Methodology	19
5 – Results	23
6 – Discussion	25
7 – Conclusion	27
Appendix	37

Chapter 1 – Writing a thesis

Writing a report is an exercise that involves both **content and form**. In this document, we aim to simplify the formatting aspect without making any assumptions about the content, specifically in the context of the ISC program¹.

1.1 – The content of a thesis

The general structure of a bachelor thesis typically includes the following sections:

1. **Abstract:** A concise summary of the thesis, including the research question, methodology, results, and conclusions.
2. **Résumé:** A summary of the thesis in French.
3. **Acknowledgements:** (Optional) A section to thank those who supported the work.
4. **Table of Contents:** An organized listing of chapters and sections.
5. **Introduction:** Presents the background, motivation, objectives, and scope and plan of the thesis.
6. **State of the Art / Literature Review:** Reviews existing research and situates the thesis within the academic context. If salient in your work.
7. **Methodology:** Describes the methods, materials, and procedures used in the research / thesis.
8. **Results:** Presents the findings of the research, often with tables, figures, and analysis.
9. **Discussion:** Interprets the results, discusses implications, and relates findings to the research question.
10. **Conclusion:** Summarizes the main findings, contributions, and suggests future work.
11. **References / Bibliography:** Lists all sources cited in the thesis.
12. **Appendices:** (Optional) Contains supplementary material such as raw data, code, or additional explanations.

This structure may vary depending on the field of study, but these elements are commonly found in most bachelor theses. They are compulsory for the *ISC Bachelor thesis*.

¹Here is how to add a footnote <https://isc.hevs.ch>

Chapter 2 – Introduction

Over the past few decades, society has been sensitized and slowly became more aware of significant problems that we are likely to face in the coming years.

Climate change and other environmental issues arise as a result of human-driven activities.

Scientists have monitored this matter and proposed various frameworks to address and mitigate these problems. In Switzerland, these different frameworks are implemented in the legislation and guidelines (at federal and canton levels) to steer the country towards a more sustainable future.

Municipalities in Switzerland are required to submit an energy planning document which outlines their future strategies to comply with those directives while also considering the characteristics of their energetical landscape.

These different properties can be quantified and analyzed through the use of a very valuable resource: data. Data is emitted by various sources ; sensors, energy models or citizen records for e.g. all yield datapoints that help us assess different indicators we are willing to measure against our municipality. These indicators, *in fine*, help us evaluate our progress towards energy-related goals.

Over the past two years, Artificial Intelligence (AI) has rapidly transformed our habits when interacting with information.

Large Language Models (LLM) allow users to interact with these systems in natural language facilitating the interface between humans and *machines*. They can provide insights into vast amounts of data at speed and scales which are beyond our capabilities.

This work tackles this exact problem that is the implementation of a solution which assists users into energy planning for a municipality.

This complex problem is approached by leveraging the power of *specialized* AIs which each offer expertise into a variety of domains and are coordinated using an *orchestration* AI to provide a solution. The expertise of the user interacting with the system tailors this solution to the specificities of the municipality.

The key steps in the engineering of this implementation are identifying the key information and datasources that are relevant to this process, structure the different AIs into an architecture whose components and interfaces are well-defined and ultimately implementing the solution.

The main objective of this work is to investigate how effective and reliable such a solution is and to assess its strengths and weaknesses. Additional goals are also outlined:

- Defining the important information in assisting user decision making.
- Understanding which datasources are available and relevant for this decision making.
- Structuring the decision by using an orchestration AI and many specialized AIs.
- Training specialized AIs on specialized datasets.
- Simplifying the user interface by handling the communication between the orchestration AI and the specialized AIs.

The project is scoped to municipalities within the canton of Valais/Wallis and strictly relies on publicly available data. Certain measures are taken to ensure the privacy and security of data that would not be of public order considering the future implementation of extra datasources.

The solution is designed to offer a user-friendly interface from which users can interact with the system in a conversational manner and visualize a map of the municipality with different layers.

User behaviour is analyzed to takeaway user preferences which gradually adapt the answers to better meet the user expectations.

TODO: provide a brief overview of the structure of the thesis (plan), add reference to extra scope in methodology?

Chapter 3 – State of the Art

Large language models are highly effective tools for natural language processing and offer various opportunities to enhance our day-to-day tasks and workflows. Ever since they have been introduced to the public, they have been adopted across a wide range of fields and applications.

TODO: citer proprement

The AI Institute at ITMO University published in 2025 a paper titled *LLM Agents for Smart City Management: Enhancing Decision Support Through Multi-Agent AI Systems*. The study examines how the natural language processing strengths of LLMs, combined with the distributed problem-solving abilities of multi-agent systems, can enhance urban decision-making processes.

The research focused on the testing of three hypotheses: (1) evaluating the capability of LLM agents to effectively route and process diverse urban queries against existing urban information systems, (2) the effectiveness of Retrieval-Augmented Generation (RAG) technology in improving response accuracy when working with local knowledge and regulations and (3) the impact of integrating LLM agents with existing urban information-systems - increasing efficiency and decreasing the decision making process time.

Their proposed solution was tested against 150 question-answer pairs and used St. Petersburg's Digital Urban Platform as a testbed. The testing dataset was curated and built by a group of human experts such as specialists in urban data analysis, GIS specialists, and urban architects.

They then evaluated different configurations of LLM agents and state-of-the-art models against two primary metrics: G-eval and answer relevance. The G-eval metric provides greater compliance with human requirements as it uses LLMs to evaluate answers from other LLMs based on custom user criteria. These criteria can for e.g. be provided as a list of rules specifying precise steps the LLM should take for evaluation, mirroring human reasoning process. The answer relevance metric, on the other hand, assesses the relevance of the answer from the LLM when compared with the correct answer provided by the experts. This process also leverages LLMs as it first extracts the different statements from the answer and then compares those to the reference answer. Both metrics are bounded between 0 and 1, where a higher value indicates better performance.

In summary, the results show greater performance when integrating the RAG technology and urban information-systems to the solution (G-eval scores of 0.68-0.74) compared to standalone LLM responses (G-eval scores of 0.30-0.38). Relevance scores, on the other hand, remain high whatever the configuration as they are inherently designed to produce semantically relevant text. They also concluded that this research proved practical real-world city management application as it enables efficient processing of urban planning tasks while maintaining high relevance in responses and shortening task completion time from days to hours.

The ITMO study presents a research-driven implementation of LLM agents focusing on decision support through integration with urban data platforms which curate and process urban data to provide insights and recommendations for urban planning and management. Rather than relying on these large-scale platforms, the present thesis explores the potential of leveraging publicly available data from federal and cantonal sources while also considering the interface with municipal archives and residents' files. This work strongly values the user experience with efforts in enhancing conversational interactions through preference-driven reporting and improving the clarity and quality of reported decisions. It neither serves as a continuation nor a re-implementation of the ITMO study, but rather represents an independent application of AI agents to a related use case specifically adapted to the context of a Bachelor's thesis and shaped by my practical implementation choices and problem-solving approach. Any solution designed and implemented around similar goals and data-related constraints, regardless of the specificities of the use case, may result in an architecture that is somewhat similar.

As we forget about the use case and consider more generic research on the matter, we encounter publications that rather focus on the optimization of various aspects of AI agents, such as their scalability, efficiency, and robustness. While this work tackles some of these challenges, it does not incorporate major research efforts into these areas.

TODO: citer aino différemment? Another AI-centric solution relevant to this use case is [aino](#), described on their homepage as an *AI GIS Analyst for Urban planning teams*. Developed and marketed in the United States, it is a commercial business solution offering a platform where an AI analyzes sites and provides visual insights from simple questions. This solution inspired me into a few design and user experience improvements in my work as no implementation details are provided.

These two points of view position this work within the fast-changing field of AI-driven solutions for urban planning and beyond.

Chapter 4 – Methodology

TODO: faire des sous chapitres

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus terminari summam voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in voluptate aut a voluptate discedere. Nam cum ignorance rerum bonarum et malarum maxime hominum vita vexetur, ob eumque errorem et voluptatibus maximis saepe priventur et durissimis animi doloribus torqueantur, sapientia est adhibenda, quae et terroribus cupiditatibusque detractis et omnium falsarum opinionum temeritate derepta certissimam se nobis ducem praebeat ad voluptatem. Sapientia enim est una, quae maestitiam pellat ex animis, quae nos exhorrescere metu non sinat. Qua praeceptrice in tranquillitate vivi potest omnium cupiditatum ardore restincto. Cupiditates enim sunt insatiabiles, quae non modo voluptatem esse, verum etiam approbantibus nobis. Sic enim ab Epicuro reprehensa et correcta permulta. Nunc dicam de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibi, si mihi probabis ea, quae dicta sunt ab iis quos probamus, eisque nostrum iudicium et nostrum scribendi ordinem adiungimus, quid habent, cur Graeca anteponant iis, quae et a formidinum terrore vindicet et ipsius fortunae modice ferre doceat iniurias et omnis monstret vias, quae ad amicos pertinerent, negarent esse per se ipsam causam non multo maiores esse et

voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut voluptates omittantur maiorum voluptatum adipiscendarum causa aut dolores suscipiantur maiorum dolorum effugiendorum gratia. Sed de clarorum hominum factis illustribus et gloriosis satis hoc loco dictum sit. Erit enim iam de omnium virtutum cursu ad voluptatem proprius disserendi locus. Nunc autem explicabo, voluptas ipsa quae qualisque sit, ut tollatur error omnis imperitorum intellegaturque ea, quae voluptaria, delicata, mollis habeatur disciplina, quam gravis, quam continens, quam severa sit. Non enim hanc solam sequimur, quae suavitate aliqua naturam ipsam movet et cum iucunditate quadam.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primum cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus terminari summam voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in voluptate aut a voluptate discedere. Nam cum ignorance rerum bonarum et malarum maxime hominum vita vexetur, ob eumque errorem et voluptatibus maximis saepe priventur et durissimis animi doloribus torqueantur, sapientia est adhibenda, quae et terroribus cupiditatusque detractis et omnium falsarum opinionum temeritate derepta certissimam se nobis ducem praebeat ad voluptatem. Sapientia enim est una, quae maestitiam pellat ex animis, quae nos exhorrescere metu non sinat. Qua praeceptrice in tranquillitate vivi potest omnium cupiditatum ardore restincto. Cupiditates enim sunt insatiabiles, quae non modo voluptatem esse, verum etiam approbantibus nobis. Sic enim ab Epicuro reprehensa et correcta permulta. Nunc dicam de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibi, si mihi probabis ea, quae dicta sunt ab iis

quos probamus, eisque nostrum iudicium et nostrum scribendi ordinem adiungimus, quid habent, cur Graeca anteponant iis, quae et a formidinum terrore vindicet et ipsius fortunae modice ferre doceat iniurias et omnis monstret vias, quae ad amicos pertinerent, negarent esse per se ipsam causam non multo maiores esse et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus, ut aut voluptates omittantur maiorum voluptatum adipiscendarum causa aut dolores suscipiantur maiorum dolorum effugiendorum gratia. Sed de clarorum hominum factis illustribus et gloriosis satis hoc loco dictum sit. Erit enim iam de omnium virtutum cursu ad voluptatem proprius disserendi locus. Nunc autem explicabo, voluptas ipsa quae qualisque sit, ut tollatur error omnis imperitorum intellegaturque ea, quae voluptaria, delicata, mollis habeatur disciplina, quam gravis, quam continens, quam severa sit. Non enim hanc solam sequimur, quae suavitate aliqua naturam ipsam movet et cum iucunditate quadam.

Chapter 5 – Results

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeo, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus terminari summam voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in voluptate aut a voluptate discedere. Nam cum ignoratione rerum bonarum et malarum maxime hominum vita vexetur, ob eumque errorem et voluptatibus maximis saepe priventur et durissimis animi doloribus torqueantur, sapientia est adhibenda, quae et terroribus cupiditatusque detractis et omnium falsarum opinionum temeritate derepta certissimam se nobis ducem praebeat ad voluptatem. Sapientia enim est una, quae maestitiam pellat ex animis, quae nos exhorrescere metu non sinat. Qua praeceptrice in tranquillitate vivi potest omnium cupiditatum ardore restincto. Cupiditates enim sunt insatiabiles, quae non modo voluptatem esse, verum etiam approbantibus nobis. Sic enim ab Epicuro reprehensa et correcta permulta. Nunc dicam de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibi, si mihi probabis ea, quae dicta sunt ab iis quos probamus, eisque nostrum iudicium et nostrum scribendi ordinem adiungimus, quid habent, cur Graeca anteponant iis, quae et a formidinum terrore vindicet et ipsius fortunae modice ferre doceat iniurias et omnis monstret vias, quae ad amicos pertinerent, negarent esse per se ipsam causam non multo maiores esse et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus,

ut aut voluptates omittantur maiorum voluptatum adipiscendarum causa aut dolores suscipiantur maiorum dolorum effugiendorum gratia. Sed de clarorum hominum factis illustribus et gloriosis satis hoc loco dictum sit. Erit enim iam de omnium virtutum cursu ad voluptatem proprius disserendi locus. Nunc autem explicabo, voluptas ipsa quae qualisque sit, ut tollatur error omnis imperitorum intellegaturque ea, quae voluptaria, delicata, mollis habeatur disciplina, quam gravis, quam continens, quam severa sit. Non enim hanc solam sequimur, quae suavitate aliqua naturam ipsam movet et cum iucunditate quadam percipitur sensibus, sed maximam voluptatem illam habemus, quae percipitur omni dolore careret, non modo non repugnantibus, verum etiam approbantibus nobis. Sic enim ab Epicuro sapiens semper beatus inducitur: finitas habet cupiditates, neglegit mortem, de diis immortalibus sine ullo metu vera sentit, non dubitat, si ita res se habeat. Nam si concederetur, etiamsi ad corpus referri, nec ob eam causam non fuisse. – Torquem detraxit hosti. – Et quidem se textit, ne interiret. – At magnum periculum adiit. – In oculis quidem exercitus. – Quid ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo et gravissimas res consilio ipsius et ratione administrari neque maiorem voluptatem ex infinito tempore aetatis percipi posse, quam ex hoc facillime perspici potest: Constituamus aliquem magnis, multis, perpetuis fruentem et animo et attento intuemur, tum fit ut aegritudo.

Chapter 6 – Discussion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turma omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeo, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus terminari summam voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in voluptate aut a voluptate discedere. Nam cum ignoratione rerum bonarum et malarum maxime hominum vita vexetur, ob eumque errorem et voluptatibus maximis saepe priventur et durissimis animi doloribus torqueantur, sapientia est adhibenda, quae et terroribus cupiditatumque detractis et omnium falsarum opinionum temeritate derepta certissimam se nobis ducem praebeat ad voluptatem. Sapientia enim est una, quae maestitiam pellat ex animis, quae nos exhorrescere metu non sinat. Qua praeceptrice in tranquillitate vivi potest omnium cupiditatum ardore restincto. Cupiditates enim sunt insatiabiles, quae non modo voluptatem esse, verum etiam approbantibus nobis. Sic enim ab Epicuro reprehensa et correcta permulta. Nunc dicam de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibi, si mihi probabis ea, quae dicta sunt ab iis quos probamus, eisque nostrum iudicium et nostrum scribendi ordinem adiungimus, quid habent, cur Graeca anteponant iis, quae et a formidinum terrore vindicet et ipsius fortunae modice ferre doceat iniurias et omnis monstret vias, quae ad amicos pertinerent, negarent esse per se ipsam causam non multo maiores esse et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus,

ut aut voluptates omittantur maiorum voluptatum adipiscendarum causa aut dolores suscipiantur maiorum dolorum effugiendorum gratia. Sed de clarorum hominum factis illustribus et gloriosis satis hoc loco dictum sit. Erit enim iam de omnium virtutum cursu ad voluptatem proprius disserendi locus. Nunc autem explicabo, voluptas ipsa quae qualisque sit, ut tollatur error omnis imperitorum intellegaturque ea, quae voluptaria, delicata, mollis habeatur disciplina, quam gravis, quam continens, quam severa sit. Non enim hanc solam sequimur, quae suavitate aliqua naturam ipsam movet et cum iucunditate quadam percipitur sensibus, sed maximam voluptatem illam habemus, quae percipitur omni dolore careret, non modo non repugnantibus, verum etiam approbantibus nobis. Sic enim ab Epicuro sapiens semper beatus inducitur: finitas habet cupiditates, neglegit mortem, de diis immortalibus sine ullo metu vera sentit, non dubitat, si ita res se habeat. Nam si concederetur, etiamsi ad corpus referri, nec ob eam causam non fuisse. – Torquem detraxit hosti. – Et quidem se textit, ne interiret. – At magnum periculum adiit. – In oculis quidem exercitus. – Quid ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo et gravissimas res consilio ipsius et ratione administrari neque maiorem voluptatem ex infinito tempore aetatis percipi posse, quam ex hoc facillime perspici potest: Constituamus aliquem magnis, multis, perpetuis fruenter et animo et attento intuemur, tum fit ut aegritudo sequatur, si illa mala sint, laetitia, si bona. O praeclaram beate vivendi et apertam et simplicem et directam viam! Cum enim certe nihil homini possit melius esse quam Graecam. Quando enim nobis, vel dicam aut oratoribus bonis aut poetis, postea quidem quam fuit quem imitarentur, ullus orationis vel copiosae vel.

Chapter 7 – Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum defuturum, quas natura non depravata desiderat. Et quem ad me accedis, saluto: 'chaere,' inquam, 'Tite!' lictores, turba omnis chorusque: 'chaere, Tite!' hinc hostis mi Albucius, hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeo, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti legant. Quis enim tam inimicus paene nomini Romano est, qui Ennii Medeam aut Antiopam Pacuvii spernat aut reiciat, quod se isdem Euripidis fabulis delectari dicat, Latinas litteras oderit? Synephebos ego, inquit, potius Caecilii aut Andriam Terentii quam utramque Menandri legam? A quibus tantum dissentio, ut, cum Sophocles vel optime scripserit Electram, tamen male conversam Atilii mihi legendam putem, de quo Lucilius: 'ferreum scriptorem', verum, opinor, scriptorem tamen, ut legendus sit. Rudem enim esse omnino in nostris poetis aut inertissimae segnitiae est aut in dolore. Omnis autem privatione doloris putat Epicurus terminari summam voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in voluptate aut a voluptate discedere. Nam cum ignoratione rerum bonarum et malarum maxime hominum vita vexetur, ob eumque errorem et voluptatibus maximis saepe priventur et durissimis animi doloribus torqueantur, sapientia est adhibenda, quae et terroribus cupiditatumque detractis et omnium falsarum opinionum temeritate derepta certissimam se nobis ducem praebeat ad voluptatem. Sapientia enim est una, quae maestitiam pellat ex animis, quae nos exhorrescere metu non sinat. Qua praeceptrice in tranquillitate vivi potest omnium cupiditatum ardore restincto. Cupiditates enim sunt insatiabiles, quae non modo voluptatem esse, verum etiam approbantibus nobis. Sic enim ab Epicuro reprehensa et correcta permulta. Nunc dicam de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibi, si mihi probabis ea, quae dicta sunt ab iis quos probamus, eisque nostrum iudicium et nostrum scribendi ordinem adiungimus, quid habent, cur Graeca anteponant iis, quae et a formidinum terrore vindicet et ipsius fortunae modice ferre doceat iniurias et omnis monstret vias, quae ad amicos pertinerent, negarent esse per se ipsam causam non multo maiores esse et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum hic tenetur a sapiente delectus,

ut aut voluptates omittantur maiorum voluptatum adipiscendarum causa aut dolores suscipiantur maiorum dolorum effugiendorum gratia. Sed de clarorum hominum factis illustribus et gloriosis satis hoc loco dictum sit. Erit enim iam de omnium virtutum cursu ad voluptatem proprius disserendi locus. Nunc autem explicabo, voluptas ipsa quae qualisque sit, ut tollatur error omnis imperitorum intellegaturque ea, quae voluptaria, delicata, mollis habeatur disciplina, quam gravis, quam continens, quam severa sit. Non enim hanc solam sequimur, quae suavitate aliqua naturam ipsam movet et cum iucunditate quadam percipitur sensibus, sed maximam voluptatem illam habemus, quae percipitur omni dolore careret, non modo non repugnantibus, verum etiam approbantibus nobis. Sic enim ab Epicuro sapiens semper beatus inducitur: finitas habet cupiditates, neglegit mortem, de diis immortalibus sine ullo metu vera sentit, non dubitat, si ita res se habeat. Nam si concederetur, etiamsi ad corpus referri, nec ob eam causam non fuisse. – Torquem detraxit hosti. – Et quidem se texit, ne interiret. – At magnum periculum adiit. – In oculis quidem exercitus. – Quid ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo et gravissimas res consilio ipsius et ratione administrari neque maiorem voluptatem ex infinito tempore aetatis percipi posse, quam ex hoc facillime perspicitur potest: Constituamus aliquem magnis, multis, perpetuis fruente et animo et attento intuemur, tum fit ut aegritudo sequatur, si illa mala sint, laetitia, si bona. O praeclaram beate vivendi et apertam et simplicem et directam viam! Cum enim certe nihil homini possit melius esse quam Graecam. Quando enim nobis, vel dicam aut oratoribus bonis aut poetis, postea quidem quam fuit quem imitarentur, ullus orationis vel copiosae vel elegantis ornatus defuit? Ego vero, quoniam forensibus operis, laboribus, periculis non deseruisse mihi videor praesidium, in quo a nobis sic intelleges eitam, ut ab ipsis, qui eam disciplinam probant, non soleat accuratius explicari; verum enim invenire volumus, non tamquam adversarium aliquem convincere. Accurate autem quondam a L. Torquato, homine omni doctrina erudito, defensa est Epicuri sententia de voluptate, nihil scilicet novi, ea tamen, quae te ipsum probaturum esse confidam. Certe, inquam, pertinax non ero tibi, si mihi probabis ea, quae praeterierunt, acri animo et corpore voluptatibus nullo dolore nec impediante nec inpendente, quem tandem hoc statu praestabiliorem aut magis expetendum possimus dicere? Inesse enim necesse est effici, ut sapiens solum amputata circumcisaque inanitate omni et errore naturae finibus contentus sine aegritudine possit et sine metu degendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo et gravissimas res consilio ipsius et ratione administrari neque maiorem voluptatem ex infinito tempore aetatis percipi posse, quam ex hoc facillime perspicitur potest: Constituamus aliquem magnis, multis, perpetuis fruente et animo et corpore voluptatibus nullo dolore nec impediante nec inpendente, quem tandem hoc statu praestabiliorem aut magis expetendum possimus dicere? Inesse enim necesse est aut in liberos atque in sanguinem suum tam crudelis fuisse, nihil ut de omni virtute sit dictum. Sed similia fere dici possunt. Ut enim virtutes, de quibus neque depravate iudicant neque corrupte, nonne ei maximam gratiam habere debemus, qui hac exaudita quasi voce naturae sic eam firme graviterque comprehenderit, ut omnes bene sanos ad iustitiam, aequitatem, fidem, neque homini infanti aut inpotenti.

Bibliography

- [1] “Learn the basics.” [Online]. Available: <https://langchain-ai.github.io/langgraph/tutorials/introduction/#setup>
- [2] “Run models locally | 🦜🔗 LangChain.” [Online]. Available: https://python.langchain.com/docs/how_to/local_llms/
- [3] “ChatOllama — 🦜🔗 LangChain documentation.” [Online]. Available: https://python.langchain.com/api_reference/ollama/chat_models/langchain_ollama.chat_models.ChatOllama.html
- [4] “tools — 🦜🔗 LangChain documentation.” [Online]. Available: https://python.langchain.com/api_reference/community/tools.html
- [5] “Machine learning as a surrogate model for EnergyPLAN: Speeding up energy system optimization at the country level - ScienceDirect.” [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S036054422402509X>
- [6] “Neural Networks with Micrograd.” [Online]. Available: <https://marimo.io/p/@marimo/micrograd>
- [7] “Free Online Gantt Chart Software.” [Online]. Available: <https://www.onlinegantt.com/#/gantt>
- [8] “Renuo – DeepSeek's Disruptive AI: How It Sets the Stage for Agentic AI.” [Online]. Available: https://www.renuo.ch/en/blog/agentic_ai
- [9] “langchain-core: 0.3.59 — 🦜🔗 LangChain documentation.” [Online]. Available: https://python.langchain.com/api_reference/core/index.html
- [10] “[2210.03629] ReAct: Synergizing Reasoning and Acting in Language Models.” [Online]. Available: <https://arxiv.org/abs/2210.03629>
- [11] “Introducing the LangGraph Functional API.” [Online]. Available: <https://blog.langchain.dev/introducing-the-langgraph-functional-api/>
- [12] “Introduction | 🦜🔗 LangChain.” [Online]. Available: <https://python.langchain.com/docs/introduction/>
- [13] “Monitoring data spaces in Switzerland.” [Online]. Available: https://www.bk.admin.ch/bk/en/home/digitale-transformation-ikt-lenkung/datenokosystem_schweiz/monitoring-datenraume.html
- [14] “OSeMOSYS - Home.” [Online]. Available: <http://www.osemosys.org/>
- [15] “(PDF) Robust Optimization for Strategic Energy Planning.” [Online]. Available: https://www.researchgate.net/publication/290496588_Robust_Optimization_for_Strategic_Energy_Planning
- [16] “EF-LLM: Energy Forecasting LLM with AI-assisted Automation, Enhanced Sparse Prediction, Hallucination Detection.” [Online]. Available: <https://arxiv.org/html/2411.00852v2>
- [17] “Building Effective AI Agents \backslash Anthropic.” [Online]. Available: <https://www.anthropic.com/engineering/building-effective-agents>
- [18] “LLM Agents for Smart City Management: Enhancing Decision Support Through Multi-Agent AI Systems. | EBSCOhost.” [Online]. Available: https://openurl.ebsco.com/EPDB%3Agcd%3A9%3A17382313/detailv2?sid=ebsco%3Aplink%3Ascholar&id=ebsco%3Agcd%3A183336541&crl=c&link_origin=scholar.google.com
- [19] “LLM Agents for Smart City Management: Enhancing Decision Support Through Multi-Agent AI Systems - EBSCO.” [Online]. Available: <https://research.ebsco.com/c/ylm4lv/viewer/pdf/nmzjt4u4hf?modal=download>
- [20] “OpenStreetMap.” [Online]. Available: <https://www.openstreetmap.org/#map=8/46.825/8.224>
- [21] “Your first project · Aino.” [Online]. Available: <https://beta.aino.world/project/323719>

- [22] “ITMO-NSS-team/llm-agents-for-smartcities-paper: Code and data for "LLM Agents for Smart City Management: Enhancing Decision Support through Multi-Agent AI Systems".” [Online]. Available: <https://github.com/ITMO-NSS-team/llm-agents-for-smartcities-paper>
- [23] “Maps of Switzerland - Swiss Confederation - map.geo.admin.ch.” [Online]. Available: https://map.geo.admin.ch/#/?map?lang=en%C2%A2er=2660000,1190000&z=1&topic=ech&layers=ch.swisstopo.zeitreihen@year=1864,f;ch.bfs.gebaeude_wohnungs_register,f;ch.bav.haltestellen-oev,f;ch.swisstopo.swisstm3d-wanderwege,f;ch.vbs.schiessanzeigen,f;ch.astra.wanderland-sperrungen_umleitungen,f&bgLayer=ch.swisstopo.pixelkarte-farbe
- [24] “Welcome to GeoAdmin API's documentation! — GeoAdmin API 3.0 documentation.” [Online]. Available: <https://api3.geo.admin.ch/index.html>
- [25] “RS 730.1 - Loi sur l'énergie - Canton du Valais - Recueil de la législation.” [Online]. Available: https://lex.vs.ch/app/fr/texts_of_law/730.1
- [26] “Plan directeur cantonal 2019 - - vs.ch.” [Online]. Available: <https://www.vs.ch/web/sdt/plan-directeur-cantonal-2019>
- [27] “Redis as a vector database quick start guide | Docs.” [Online]. Available: <https://redis.io/docs/latest/develop/get-started/vector-database/>
- [28] “geocat.ch.” [Online]. Available: <https://www.geocat.ch/datahub/search>
- [29] “geocat.ch.” [Online]. Available: <https://www.geocat.ch/geonetwork/srv/eng/catalog.search#/home>
- [30] “https://api3.geo.admin.ch/rest/services/api/SearchServer?searchText=solaire&type=layers&lang=fr.” [Online]. Available: <https://api3.geo.admin.ch/rest/services/api/SearchServer?searchText=solaire&type=layers&lang=fr>
- [31] “Guide de planification énergétique territoriale.” [Online]. Available: <https://www.local-energy.swiss/fr/infobox/raeumliche-energieplanung.html#/>
- [32] “(2019) Vision 2060 et objectifs 2035.pdf.” [Online]. Available: <https://www.vs.ch/documents/16739272/16831569/%282019%29+Vision+2060+et+objectifs+2035.pdf/9d08113e-7ad6-43a9-ac0f-0a89f857ffd7?t=1556524821961&v=1.0>
- [33] “Guide_planification_énergétique_24.11.2023.pdf.” [Online]. Available: https://www.vd.ch/fileadmin/user_upload/themes/environnement/energie/fichiers_pdf/Guide_planification_%C3%A9nergie%C3%A9tique_24.11.2023.pdf
- [34] “Géodonnées - Service de la géoinformation - vs.ch.” [Online]. Available: <https://www.vs.ch/web/egeo/geodonnees>
- [35] “Cartes interactives - Service de la géoinformation - vs.ch.” [Online]. Available: <https://www.vs.ch/web/egeo/cartes-interactives>
- [36] “Ordonnance sur l'énergie (OcEne).pdf.” [Online]. Available: <https://www.vs.ch/documents/16739272/21048558/Ordonnance+sur+l%27%C3%A9nergie+%28OcEne%29.pdf/0d27688c-b8a9-e087-e51c-d4d92b477468?t=1718958084754>
- [37] “RS 705.1 - Loi sur les constructions - Canton du Valais - Recueil de la législation.” [Online]. Available: https://lex.vs.ch/app/fr/texts_of_law/705.1
- [38] “RegBL | Registre fédéral des bâtiments et des logements.” [Online]. Available: <https://www.housing-stat.ch/fr/madd/public.html#webservices>
- [39] “RegBL | Registre fédéral des bâtiments et des logements.” [Online]. Available: <https://www.housing-stat.ch/fr/query/egid.html?egid=20>

- [40] “Fiches thématiques et articles-type - - vs.ch.” [Online]. Available: <https://www.vs.ch/fr/web/sdt/fiches-thematiques>
- [41] “geoadmin/ol3: OpenLayers 3.” [Online]. Available: <https://github.com/geoadmin/ol3>
- [42] “Web Map Tiling Services WMTS: Services et données disponibles.” [Online]. Available: <https://www.geo.admin.ch/fr/wmts-services-et-donnees-disponibles>
- [43] “Interface de programmation API : simple et performante.” [Online]. Available: <https://www.geo.admin.ch/fr/interface-de-programmation-api>
- [44] “geocat.ch.” [Online]. Available: <https://www.geocat.ch/geonetwork/srv/fre/catalog.search#/home>
- [45] “Solarrechner - myky - Kennen Sie Ihr Solarpotenzial schon?” [Online]. Available: https://solarrechner.myky.ch/?lang=fr-ch&gad_source=1&gad_campaignid=22096885391&gclid=Cj0KCQjwoZbBBhDCARIsAOqMEZV8ZiqOvL0kRs8RyPDn37mnA5ZgZRIkYU7F9tc7nHFLSINlxZmhqpEaAgdmEALw_wcB
- [46] “Get Parcel from address.” [Online]. Available: <https://groups.google.com/g/geoadmin-api/c/O1KGqNJoX4Y/m/nhgOrShEBwAJ>
- [47] “geoadmin/web-mapviewer: The viewer application for maps and geodata.” [Online]. Available: <https://github.com/geoadmin/web-mapviewer>
- [48] “Swiss coordinate system - Wikipedia.” [Online]. Available: https://en.wikipedia.org/wiki/Swiss_coordinate_system
- [49] “Building for Production | Vite.” [Online]. Available: <https://vite.dev/guide/build>
- [50] “Home | react-map-gl.” [Online]. Available: <https://visgl.github.io/react-map-gl/>
- [51] “Geometry objects | ArcGIS REST APIs | ArcGIS Developers.” [Online]. Available: <https://developers.arcgis.com/rest/services-reference/enterprise/geometry-objects/>
- [52] “Catalogue des caractères.” [Online]. Available: <https://www.housing-stat.ch/catalog/fr/4.2/GWAERZH1/final>
- [53] “Conception énergie éolienne.” [Online]. Available: <https://www.are.admin.ch/are/fr/home/developpement-et-amenagement-du-territoire/strategie-et-planification/conceptions-et-plans-sectoriels/conceptions/conception-energie-eolienne.html>
- [54] “langgraph/libs/sdk-py at main · langchain-ai/langgraph.” [Online]. Available: <https://github.com/langchain-ai/langgraph/tree/main/libs/sdk-py>
- [55] “Enhancement Proposal: PydanticStreamOutputParser for Stream and Improved LLM JSON Output · langchain-ai/langchain · Discussion #19225.” [Online]. Available: <https://github.com/langchain-ai/langchain/discussions/19225>
- [56] “InMemoryVectorStore — 📘 LangChain documentation.” [Online]. Available: https://python.langchain.com/api_reference/core/vectorstores/langchain_core.vectorstores.in_memory.InMemoryVectorStore.html
- [57] “Integrate LangGraph into a React app.” [Online]. Available: https://langchain-ai.github.io/langgraph/cloud/how-tos/use_stream_react/
- [58] “Getting Started | assistant-ui.” [Online]. Available: <https://www.assistant-ui.com/docs/runtimes/langgraph>
- [59] “Using uv with FastAPI | uv.” [Online]. Available: <https://docs.astral.sh/uv/guides/integration/fastapi/#migrating-an-existing-fastapi-project>

- [60] “Use SSE streaming · langchain-ai/langgraph-fullstack-python@e74f8d5.” [Online]. Available: <https://github.com/langchain-ai/langgraph-fullstack-python/commit/e74f8d5fe9f4e9ec22b1243ca32bade5ebf05c85#diff-221b01b2a9dbfd122a46fccd82d22ba679a511f48185eb6c4f06308a151c5353>
- [61] “mem0ai/mem0: Memory for AI Agents; SOTA in AI Agent Memory; Announcing OpenMemory MCP - local and secure memory management.” [Online]. Available: <https://github.com/mem0ai/mem0>
- [62] “STAC API.” [Online]. Available: <https://www.bfe.admin.ch/bfe/en/home/supply/digitalization-and-geoinformation/programming-interfaces/stac-api.html>
- [63] “2403.04656v2.pdf.” [Online]. Available: <https://arxiv.org/pdf/2403.04656>
- [64] “Homepage | GSAP.” [Online]. Available: <https://gsap.com/>
- [65] “Rerankers and Two-Stage Retrieval | Pinecone.” [Online]. Available: <https://www.pinecone.io/learn/series/rag/rerankers/>
- [66] “sentence-transformers/all-MiniLM-L6-v2 · Hugging Face.” [Online]. Available: <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>
- [67] “Statistique de l’électricité.” [Online]. Available: <https://www.bfe.admin.ch/bfe/fr/home/approvisionnement/statistiques-et-geodonnees/statistiques-de-lenergie/statistique-de-l-electricite.html/>
- [68] “Statistiques sectorielles.” [Online]. Available: <https://www.bfe.admin.ch/bfe/fr/home/approvisionnement/statistiques-et-geodonnees/statistiques-de-lenergie/statistiques-sectorielles.html>
- [69] “6238.pdf.” [Online]. Available: https://www.uvek-gis.admin.ch/BFE/storymaps/ECH_SolarpotGemeinden/pdf/6238.pdf
- [70] “ollama-container/Singularity at main · uw-psych/ollama-container.” [Online]. Available: <https://github.com/uw-psych/ollama-container/blob/main/Singularity>
- [71] “How to load PDFs | 🦜🔗 LangChain.” [Online]. Available: https://python.langchain.com/docs/how_to/document_loader_pdf/
- [72] “vidore/colpali-v1.3 · Hugging Face.” [Online]. Available: <https://huggingface.co/vidore/colpali-v1.3>
- [73] “How to parse PDF docs for RAG.” [Online]. Available: https://cookbook.openai.com/examples/parse_pdf_docs_for_rag
- [74] “qwen2.5vl.” [Online]. Available: <https://ollama.com/library/qwen2.5vl>
- [75] “Qwen/Qwen2.5-VL-7B-Instruct · Hugging Face.” [Online]. Available: <https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>
- [76] “RetrievalTutorials/tutorials/LevelsOfTextSplitting/5_Levels_Of_Text_Splitting.ipynb at main · Full-StackRetrieval-com/RetrievalTutorials.” [Online]. Available: https://github.com/FullStackRetrieval-com/RetrievalTutorials/blob/main/tutorials/LevelsOfTextSplitting/5_Levels_Of_Text_Splitting.ipynb
- [77] “How to split Markdown by Headers | 🦜🔗 LangChain.” [Online]. Available: https://python.langchain.com/docs/how_to/markdown_header_metadata_splitter/
- [78] “Run Redis Open Source on Docker | Docs.” [Online]. Available: https://redis.io/docs/latest/operate/oss_and_stack/install/install-stack/docker/
- [79] “redis-ai-resources/python-recipes/RAG/02_langchain.ipynb at main · redis-developer/redis-ai-resources.” [Online]. Available: https://github.com/redis-developer/redis-ai-resources/blob/main/python-recipes/RAG/02_langchain.ipynb
- [80] “How-to guides | 🦜🔗 LangChain.” [Online]. Available: https://python.langchain.com/docs/how_to/#prompt-templates
- [81] “Manage memory.” [Online]. Available: <https://langchain-ai.github.io/langgraph/how-tos/memory/>

- [82] “Overview.” [Online]. Available: <https://langchain-ai.github.io/langgraph/concepts/memory/#episodic-memory>
- [83] “Overview.” [Online]. Available: <https://langchain-ai.github.io/langgraph/concepts/memory/#procedural-memory>
- [84] “ELI5: How does ChatGPT's memory actually work behind the scenes? : r/OpenAI.” [Online]. Available: https://www.reddit.com/r/OpenAI/comments/1jy3e6z/eli5_how_does_chatgpts_memory_actually_work/
- [85] “How to Extract Semantic Memories.” [Online]. Available: https://langchain-ai.github.io/langmem/guides/extract_semantic_memories/#without-storage
- [86] “Parsing PDFs for Image Retrieval in the RAG Framework | by Nvmukesh | Medium.” [Online]. Available: <https://medium.com/@nvmukesh/parsing-pdfs-for-image-retrieval-in-the-rag-framework-d1797e0843a2>
- [87] “[2407.01449] ColPali: Efficient Document Retrieval with Vision Language Models.” [Online]. Available: <https://arxiv.org/abs/2407.01449>
- [88] “Layout-Parser/layout-parser: A Unified Toolkit for Deep Learning Based Document Image Analysis.” [Online]. Available: <https://github.com/Layout-Parser/layout-parser?tab=readme-ov-file>
- [89] “vidore/colqwen2-v1.0 · Hugging Face.” [Online]. Available: <https://huggingface.co/vidore/colqwen2-v1.0>
- [90] “vidore/colqwen2.5-v0.2 · Hugging Face.” [Online]. Available: <https://huggingface.co/vidore/colqwen2.5-v0.2>
- [91] “What is a Recommendation System in Graph Databases? – Hypermode.” [Online]. Available: <https://hypermode.com/blog/recommendation-system-design>
- [92] “LLMGraphTransformer — 🦜 LangChain documentation.” [Online]. Available: https://python.langchain.com/api_reference/experimental/graph_transformers/langchain_experimental_graph_transformers.llm.LLMGraphTransformer.html
- [93] “How to construct knowledge graphs | 🦜 LangChain.” [Online]. Available: https://python.langchain.com/docs/how_to/graph_constructing/
- [94] “Tools | 🦜 LangChain.” [Online]. Available: <https://python.langchain.com/docs/concepts/tools/>
- [95] “Cross Encoder Reranker | 🦜 LangChain.” [Online]. Available: https://python.langchain.com/docs/integrations/document_transformers/cross_encoder_reranker/
- [96] “Hierarchical Topic Modeling - BERTopic.” [Online]. Available: https://maartengr.github.io/BERTopic/getting_started/hierarchicaltopics/hierarchicaltopics.html
- [97] “NOGA 2025 | KUBB - Application d'aide à la codification.” [Online]. Available: <https://www.kubb-tool.bfs.admin.ch/fr/noga/2025>
- [98] “Effects of Prompt Length on Domain-specific Tasks for Large Language Models.” [Online]. Available: <https://arxiv.org/html/2502.14255v1#S5>
- [99] “Faithfulness - Ragas.” [Online]. Available: https://docs.ragas.io/en/stable/concepts/metrics/available_metrics/faithfulness/
- [100] “YOU CAN EXTRACT REASONING FROM R1 AND PASS IT ONTO ANY MODEL : r/LocalLLaMA.” [Online]. Available: https://www.reddit.com/r/LocalLLaMA/comments/1i73x81/you_can_extract_reasoning_from_r1_and_pass_it/

- [101] “Performance Impact of Scaling a 70B Model Across Multiple A100 GPUs and Further Speed Optimization · Issue #7648 · ollama/ollama.” [Online]. Available: <https://github.com/ollama/ollama/issues/7648#issuecomment-2473561990>
- [102] “python-dotenv·PyPI.” [Online]. Available: <https://pypi.org/project/python-dotenv/>
- [103] “Coefficient of variation - Wikipedia.” [Online]. Available: https://en.wikipedia.org/wiki/Coefficient_of_variation
- [104] “Quartile coefficient of dispersion - Wikipedia.” [Online]. Available: https://en.wikipedia.org/wiki/Quartile_coefficient_of_dispersion
- [105] “Indicateurs énergétiques communaux - - vs.ch.” [Online]. Available: <https://www.vs.ch/web/energie/indicateurs-energetiques-communaux>
- [106] “Algorithms for calculating variance - Wikipedia.” [Online]. Available: https://en.wikipedia.org/wiki/Algorithms_for_calculating_variance#Welford's_online_algorithm

Appendix

Table of Acronyms

AI :	Artificial Intelligence
LLM :	Large Language Model
RAG :	Retrieval-Augmented Generation

List of Figures

Code Appendix

```

1 class Cons[+A](hd: A, tl: => MyStream[A]) extends MyStream[A] {
2   override def isEmpty: Boolean = false
3   override val head: A = hd
4   override lazy val tail: MyStream[A] = tl
5   override def #::[B >: A](elem: B): MyStream[B] =
6     new Cons[B](elem, this)
7
8   override def ++[B >: A](anotherStream: => MyStream[B]): MyStream[B] =
9     new Cons[B](head, tail ++ anotherStream)
10
11  override def foreach(f: A => Unit): Unit = {
12    f(head)
13    tail.foreach(f)
14  }
15 }

```

Listing 1 - Code included from the file example.scala

```

1 def mergeSort(arr):
2   if len(arr) <= 1:
3     return arr
4
5   mid = len(arr) // 2
6   leftHalf = arr[:mid]
7   rightHalf = arr[mid:]
8
9   sortedLeft = mergeSort(leftHalf)
10  sortedRight = mergeSort(rightHalf)
11
12  return merge(sortedLeft, sortedRight)
13
14 def merge(left, right):
15   result = []
16   i = j = 0
17
18   while i < len(left) and j < len(right):
19     if left[i] < right[j]:
20       result.append(left[i])
21       i += 1
22     else:
23       result.append(right[j])
24       j += 1
25
26   result.extend(left[i:])
27   result.extend(right[j:])
28
29   return result
30
31 mylist = [3, 7, 6, -10, 15, 23.5, 55, -13]
32 mysortedlist = mergeSort(mylist)
33 print("Sorted array:", mysortedlist)

```

Listing 2 - Second code included from the file example.scala

```

1 class Cons[+A](hd: A, tl: => MyStream[A]) extends MyStream[A] {
2   override def isEmpty: Boolean = false
3   override val head: A = hd
4   override lazy val tail: MyStream[A] = tl
5   override def #::[B >: A](elem: B): MyStream[B] =
6     new Cons[B](elem, this)
7
8   override def ++[B >: A](anotherStream: => MyStream[B]): MyStream[B] =
9     new Cons[B](head, tail ++ anotherStream)

```

```
10  
11 override def foreach(f: A => Unit): Unit = {  
12     f(head)  
13     tail.foreach(f)  
14 }  
15 }
```

Listing 3 - Second code included from the file example.scala